1. INTRODUCTION

1.1 Project Overview

The cloud computing sector is expanding at a breakneck pace, but many data centers and server farms still use traditional load balancing techniques that ignore the physical thermal conditions of their hardware. With hardware running at higher capacities, the need for a robust, thermal-sensitive routing platform is becoming more and more prevalent.

To address this need, we built a tool called the Thermal-Aware Load Balancer. This project is a simple, easy-to-use web application that can be run on a PC or on a laptop. It interacts with backend server clusters to constantly monitor hardware sensors. Administrators can view real-time graphs, see ahead-of-time deep learning temperature predictions, and monitor a dynamic routing system.

Instead of mindlessly routing traffic, our system assigns a "thermal score" based on ahead-of-time predictions for CPU temperature, CPU usage, power draw, and GPU temperature. Traffic is routed to the coolest, healthiest servers available, linking hardware monitoring with application-level traffic routing.

The main purpose of this project is to make data center management proactive and efficient. By providing all the necessary prediction and routing tools in one place, administrators can focus on being more productive, while the system handles protecting hardware from thermal damage.

1.2 Need Analysis

The landscape of cloud infrastructure is rapidly evolving, and as it does, data centers are facing a number of challenges when it comes to load balancing with respect to temperature. We hope to solve these issues, which can have real-world implications:

Increased Hardware Costs: Excessive heat degrades hardware over time, and most existing load balancers have no concept of temperature, so if hot servers are still under high load (with respect to CPU), traffic will be routed to them, potentially causing early hardware failure.

Unpredictable Delays: If a server gets too hot, it may slow down its processing to compensate. This is called thermal throttling. In many cases, thermal throttling causes unpredictable delays to end users.

Too Much Energy Spent Cooling: It is common for data center cooling systems to over-compensate for hot spots, resulting in a waste of electricity. Most traditional load balancing does not take this into account.

No Short-Term Forecasting: Most existing monitoring tools only provide alerts when a server is already overheating. There is a need for more short-term forecasting to allow for load shifting before it becomes an issue.

We believe we can improve cloud infrastructure as a whole by solving for these issues.

1.3 Problem Definition and Scope

Cloud computing has become a crucial part of today's technology-driven world. As more and more digital services, automation, and virtualization have emerged, temperature regulation of servers has become a critical problem to solve. For data center administrators, preserving the lifetime

of the physical hardware and maximizing energy efficiency are two crucial goals; yet, a huge amount of money is spent on cooling over-usage and frequent CPU replacements.

One of the problems in this field is that conventional load balancing techniques can cause thermal imbalances and hotspots in some servers. Therefore, a smarter solution is required for these problems, and this thesis aims to exploit deep learning approaches, namely LSTM, GRU-CNN, and Transformer, for temperature prediction and hot-spot avoidance by traffic directing. In return, it would preserve the lifetime of the physical hardware and speed up the application response.

1.4 Assumptions and Constraints

We anticipate that hardware sensor monitoring will be implemented on the backend servers, via something like LibreHardwareMonitor [9], to monitor metrics such as CPU package power and GPU core temperature.

The system must have access to a reliable, low-latency internal network in order to stream WebSocket telemetry.

For optimal performance, Deep Learning inference should be carried out on a dedicated node to avoid overloading the servers running the balancing logic.

Furthermore, all backend servers must run on compatible operating systems in order to probe those sensors.

1.5 Objectives

- The key objectives of this project are:
- To design a real-time service to gather telemetry data using LibreHardwareMonitor.
- To train and evaluate Deep Learning models including but not limited to LSTM, GRU-CNN and Transformers for temperature prediction.
- To design a thermal score algorithm with a dynamic priority queue to process requests efficiently.
- To provide a better user experience for system administrators via a real-time dashboard in React.

1.6 Methodology

Here we describe the design and methodology adopted for the different modules of this project.

Backend:

Spring Boot (Java) has been chosen as the backend because with a large number of continuous telemetry streams, it is a robust, scalable solution for maintaining many WebSocket connections at once.

Prediction Service:

The deep learning models, developed in Python with TensorFlow, are exposed as a microservice. The Spring Boot backend sends rolling windows of telemetry data to this service in order to predict temperature.

Priority Queue Manager:

The Thermal Score is continuously computed using the formula

Thermal Score =

0.45 * PredictedTemp + 0.25 * CPUUsage + 0.15 * Power + 0.10 * GPUTemp + 0.05 * TargetLoad.

All servers available are prioritized accordingly, in a priority queue.

APIs:

REST APIs like /api/lb/queue are used to glue the backend services to the frontend.

Frontend:

The frontend has been written in React.js so that the dynamic nature of the webpage (for example, the live SVG charts) does not require any page reloads.

Data Collection:

A custom StressGenerator script generates the simulated, fluctuating CPU target load during the 8-hour long telemetry collection sessions, used to train the neural networks.

1.7 Project Outcomes and Deliverables

The project is intended to design a platform to cater to the needs of a data center for proactive thermal load management. The goal is to build a predictive routing service and a dependable, device-compatible web application dashboard. The project is designed to make the data about the server health easily accessible and thermal load balancing automated, making cluster management easy and hardware safe.

1.8 Novelty of Work

This proposed solution differs from available open-source load balancing solutions like NGINX and HAProxy, which don't provide predictive thermal-aware routing as a default. This solution is holistic, eliminating the need for administrators to write custom scripts for overheating prevention, which makes the platform much more powerful. Load balancing and thermal monitoring have traditionally been treated as two separate subjects. This application combines the two to provide a load balancing solution where traffic can be intelligently distributed according to the thermal state of the underlying hardware.